

COMPLEX ANALYSIS NOTES

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ABSTRACT. This is a compilation of Complex Analysis notes for the course OEC-Complex Analysis taught in the academic year 2025-26 to SY B.Tech. students at COEP Technological University, Pune.

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1. BASICS OF COMPLEX NUMBERS

Numbers of the form $x + iy$ where $x, y \in \mathbb{R}$ and $i = \sqrt{-1}$ are called Complex numbers. We denote $\mathbb{C}$ to be the set of all complex numbers. Let $z = x + iy$, then $\text{Im}(z) = y$ and $\text{Re}(z) = x$ denote the imaginary and real part of z respectively. Let $z_1 = x_1 + iy_1, z_2 = x_2 + iy_2$, then addition of $z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2)$ and the difference $z_1 - z_2$ is given by $z_1 - z_2 = (x_1 - x_2) + i(y_1 - y_2)$. The product $z_1 z_2 = (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + x_2 y_1)$. For $z = x + iy$, the complex conjugate $\bar{z} := x - iy$ and modulus $|z| := \sqrt{x^2 + y^2}$. One easily checks that $z \bar{z} = |z|^2$. Now we are ready to define division of two complex numbers. Let $z_1 = x_1 + iy_1, z_2 = x_2 + iy_2 \in \mathbb{C}$ and $z_2 \neq 0$. Then

$$\frac{z_1}{z_2} = \frac{z_1 \bar{z}_2}{z_2 \bar{z}_2} = \frac{z_1 \bar{z}_2}{|z_2|^2} = \left(\frac{x_1 x_2 + y_1 y_2}{x_2^2 + y_2^2} \right) + i \left(\frac{x_2 y_1 - x_1 y_2}{x_2^2 + y_2^2} \right).$$

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1.1. Properties of Conjugation and Modulus. The complex conjugate and modulus satisfy several fundamental properties that mirror familiar rules from real numbers while extending naturally to $\mathbb{C}$.

1.1.1. Properties of Complex Conjugation. For $z, z_1, z_2 \in \mathbb{C}$,

- $\bar{z} = z$ if and only if z is real (i.e., $\text{Im}(z) = 0$).
- $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$ and $\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$.
- $\frac{\bar{z}_1}{z_2} = \frac{\bar{z}_1}{\bar{z}_2}$ provided $z_2 \neq 0$.

Proof sketch: Let $z_j = x_j + iy_j$ for $j = 1, 2$. The first follows since $\bar{z} = x - iy = x + iy$ implies $y = 0$. Addition and multiplication conjugate componentwise by direct computation, as in the given product formula. Division holds by multiplying numerator and denominator by $\bar{z}_2 z_1$.

1.1.2. Properties of the Modulus. The modulus $|z| = \sqrt{x^2 + y^2} \geq 0$ has the following properties.

- $|z| = 0$ if and only if $z = 0$.
- $|z_1 z_2| = |z_1| |z_2|$.
- $|z_1 \pm z_2| \leq |z_1| + |z_2|$ (triangle inequality).
- $|z_1 - z_2| \geq ||z_1| - |z_2||$.
- $\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}$ provided $z_2 \neq 0$.

Example: For $z_1 = 1 + i$, $z_2 = 1 - i$, we have $|z_1| = \sqrt{2}$, $|z_2| = \sqrt{2}$, and $|z_1 z_2| = |(1 + i)(1 - i)| = |2| = 2 = \sqrt{2} \cdot \sqrt{2}$. Also, $|z_1 + z_2| = |2| = 2 \leq 2\sqrt{2}$.

2. TYPES OF SETS

In this chapter, we will discuss different type of sets that will be used in further chapters.

2.1. Open sets.

Definition 1. Let $z_0 \in \mathbb{C}$, and $\epsilon > 0$. The ϵ neighborhood of z_0 is denoted by $N_\epsilon(z_0)$ and is defined to be the set $\{z \in \mathbb{C} : |z - z_0| < \epsilon\}$. The deleted- ϵ neighborhood of z_0 is denoted by $N'_\epsilon(z_0)$ and is defined to be the set $\{z \in \mathbb{C} : |z - z_0| < \epsilon, z \neq z_0\}$.

Example 1. (1) Let $z_0 = 0, \epsilon = 1$, then

$$N_1(0) = \{z \in \mathbb{C} : |z| < 1\} = \{x + iy : x, y \in \mathbb{R}, x^2 + y^2 < 1\}.$$

(2) Let $z_0 = i, \epsilon = 2$, then

$$N_2(i) = \{z \in \mathbb{C} : |z - i| < 2\} = \{x + iy : x, y \in \mathbb{R}, x^2 + (y - 1)^2 < 2\}.$$

Definition 2. Let $S \subset \mathbb{C}$. A point $z_0 \in S$ is said to be an interior point of S if there exists $\epsilon > 0$ such that $N_\epsilon(z_0) \subset S$.

Example 2. (1) Let $S_1 = \{z \in \mathbb{C} : |z| \leq 1\}$ and $z_0 = \frac{1}{2}$. Note that

$$N_{\frac{1}{3}}\left(\frac{1}{2}\right) = \left\{z \in \mathbb{C} : \left|z - \frac{1}{2}\right| < \frac{1}{3}\right\} \subset S_1.$$

To see this, let $z \in N_{\frac{1}{3}}(\frac{1}{2})$. Then

$$|z| \leq \left|z - \frac{1}{2}\right| + \frac{1}{2} \leq \frac{1}{3} + \frac{1}{2} = \frac{5}{6} < 1.$$

This proves $\frac{1}{2}$ is a limit point for S_1 .

Definition 3. Let $S \subset \mathbb{C}$. S is open in $\mathbb{C}$ if every $z \in S$ is an interior point. Equivalently, for every $z \in S$, there exists $\epsilon > 0$ depending on z such that $N_\epsilon(z) \subset S$.

Example 3. (1) Let $S_1 = \{z \in \mathbb{C} : \operatorname{Re}(z) > 0\}$. Then S_1 is open. To see this, let $z_0 \in S_1$. Then $x_0 = \operatorname{Re}(z_0) > 0$. Consider the $\frac{x_0}{2}$ neighborhood of z_0 given by the set

$$\left\{z \in \mathbb{C} : |z - z_0| < \frac{x_0}{2}\right\}.$$

We are required to show $N_{z_0}(\frac{x_0}{2}) = \{z \in \mathbb{C} : |z - z_0| < \frac{x_0}{2}\} \subset S_1$. This is clear since $z \in N_{z_0}(\frac{x_0}{2})$, then $\operatorname{Re}(z) \geq x_0 - (\frac{x_0}{2}) = \frac{x_0}{2} > 0$.

(2) Let $S_2 = \{z \in \mathbb{C} : \operatorname{Re}(z) \geq 0\}$. Then S_2 is not open. Consider $0 \in S_2$. An ϵ neighborhood of 0 is given by $\{z \in \mathbb{C} : |z| < \epsilon\}$. Clearly $-\frac{\epsilon}{3} + i\frac{\epsilon}{3}$ lies in the neighborhood but $-\frac{\epsilon}{3} + i\frac{\epsilon}{3} \notin S_2$. Hence 0 is not an interior point.

Proposition 2.1. Let S_1, S_2 be two open sets in $\mathbb{C}$. Then $S_1 \cap S_2$ is an open set.

Proof. Let $z_0 \in S_1 \cap S_2$. Since S_1 is open, there exists $\epsilon_1 > 0$ such that $N_{\epsilon_1}(z_0) \subset S_1$. Similarly, there exists $\epsilon_2 > 0$ such that $N_{\epsilon_2}(z_0) \subset S_2$. Let $\epsilon = \min(\epsilon_1, \epsilon_2) > 0$ and consider $N_\epsilon(z_0)$. $N_\epsilon(z_0)$ is contained in $N_{\epsilon_1}(z_0)$ as $\epsilon \leq \epsilon_1$ and $N_{\epsilon_1}(z_0) \subset S_1$. Similarly $N_\epsilon(z_0) \subset N_{\epsilon_2}(z_0) \subset S_2$. Hence $N_\epsilon(z_0) \subset S_1 \cap S_2$. This shows $S_1 \cap S_2$ is open as we have shown z_0 is an interior point of $S_1 \cap S_2$. $\square$

Let I be an indexing set. Let $\{S_j\}_{j \in I}$ be a collection of open sets. Then $\bigcap_{j \in I} S_j$ may not be an open set. To see this, let $I = \mathbb{N}$ and $S_j = \{z \in \mathbb{C} : |z| < \frac{1}{j}\}$. Then $\bigcap_{j \in \mathbb{N}} S_j = \{0\}$ which is not open.

Proposition 2.2. *Let I be an indexing set. Let $\{S_j\}_{j \in I}$ be a collection of open sets. Then $\bigcup_{j \in I} S_j$ is an open set.*

Proof. Left as an exercise to the reader. □

2.2. Closed sets.

Definition 4. *Let $S \subset \mathbb{C}$. We say $z \in \mathbb{C}$ is a limit point of S if for every $\epsilon > 0$, the intersection of the deleted- ϵ neighborhood of z_0 given by $N'_\epsilon(z_0)$ and S is nonempty.*

Example 4. • Let $S_1 = \{z \in \mathbb{C} : |z| \leq 1\}$. Then 1 is a limit point of S_1 . To see this consider $N'_\epsilon(1) = \{z : |z - 1| < \epsilon, z \neq 1\}$. If $\epsilon \geq 1$, Then $\frac{1}{2} \in N'_\epsilon(1)$. If $\epsilon < 1$, then $1 - \epsilon \in N'_\epsilon(1) \cap S_1$.

- Let $S_2 = \{\frac{1}{it} : t \in \mathbb{N}\}$, where $\mathbb{N}$ is the set of natural numbers. Then $\frac{1}{2i}$ is not a limit point of S_2 . Consider $N'_\frac{1}{6}(\frac{1}{2i}) = \{z : |z - \frac{1}{2i}| < \frac{1}{6}, z \neq \frac{1}{2i}\}$. Clearly $N'_\frac{1}{6}(\frac{1}{2i}) \cap S_2 = \emptyset$. Since for $\epsilon = \frac{1}{6}$ the intersection of the deleted- $\frac{1}{6}$ neighborhood of $\frac{1}{2i}$ and S_2 is empty, $\frac{1}{2i}$ is not a limit point of S_2 . Let us now see that 0 is a limit point of S_2 . Let $\epsilon > 0$ be given. There exists $n \in \mathbb{N}$ such that $\frac{1}{n} < \epsilon$. Consider $|\frac{1}{ni}| = \frac{1}{n} < \epsilon$ which implies $\frac{1}{ni} \in N'_\epsilon(0)$.

Definition 5. *Let $S \subset \mathbb{C}$. We say S is closed if it contains all its limit points.*

Example 5. (1) Let $S_1 = \{z \in \mathbb{C} : |z| \leq 1\}$. The set of all limit points of S_1 is $\{z \in \mathbb{C} : |z| \leq 1\} \subset S_1$. Hence S_1 is closed. $S_2 = \{z \in \mathbb{C} : |z| < 1\}$. One can show that 1 (see Example 4) is a limit of S_2 . Hence S_2 is not closed as $1 \notin S_2$ but is a limit point of S_2 .

- (2) Let $S_3 = \{\frac{1}{it} : t \in \mathbb{N}\}$, then S_3 is not closed as $0 \notin S_3$. Let $S_4 = \{\frac{1}{it} : t \in \mathbb{N}\} \cup \{0\}$. Then S_4 is closed as the set of all limit points for S_3 is $\{0\}$.

Theorem 1. *Let $S \subset \mathbb{C}$ and $S^c = \mathbb{C} \setminus S$. Then S is closed if and only if S^c is open.*

Proof. Let S be closed. Suppose S^c is not open. Then there exists $z_0 \in S^c$ such that z_0 is not an interior point of S^c . By the definition of an interior point, for every $\epsilon > 0$, $N_\epsilon(z_0)$ is not a subset of S^c . This means for every $\epsilon > 0$, S^c is not open. As $z_0 \in S^c$, $N'_\epsilon(z_0) \cap S \neq \emptyset$. Hence z_0 is a limit point of S which implies $z_0 \in S$ as S is closed. However, $z_0 \in S \cap S^c$ is an impossible. Hence our assumption of S^c to be not open must be false. This proves S^c is open.

Conversely, let S^c is open. Suppose S is not closed. Then there exists a limit point z_1 of the set S such that $z_1 \in S^c$. Let $\epsilon > 0$ be given. Then $N'_\epsilon(z_1) \cap S \neq \emptyset$ which implies $N_\epsilon(z_1)$ can not be a subset of S^c . Hence z_1 is not an interior point of S^c . The assumption S^c open implies $z_1 \notin S^c$, a contradiction. Thus S must be closed. □

Proposition 2.3. *Let S_1 and S_2 be two closed sets in $\mathbb{C}$. Then $S_1 \cup S_2$ is a closed set.*

Proof. By Theorem 1, S_1^c and S_2^c are open sets in $\mathbb{C}$. By Proposition 2.1 $S_1^c \cap S_2^c$ is open. Again using Theorem 1, we see that $(S_1^c \cap S_2^c)^c = S_1 \cup S_2$ is closed, where the equality follows from Demorgan's law. □

Proposition 2.4. *Let I be an indexing set. Let $\{S_j\}_{j \in I}$ be a collection of closed sets. Then $\bigcap_{j \in I} S_j$ is an open set.*

Proof. The result follows from Proposition 2.2 and Theorem 1. □

2.3. Bounded sets.

Definition 6. *Let $S \subset \mathbb{C}$. We say S is bounded if there exists $M > 0$ such that $|z| \leq M$ for every $z \in S$. If S is not bounded, then S is said to be unbounded.*

Example 6. (1) Let $S_1 = \{z : \operatorname{Re}(z) < 1, \operatorname{Im}(z) < 2\}$. Then S_1 is bounded. To see this, let $z = x + yi \in S_1$ with $x, y \in \mathbb{R}$. Then $|z| = \sqrt{x^2 + y^2} < \sqrt{1 + 4} = \sqrt{5}$.
(2) Let $S_2 = \{z : \operatorname{Re}(z) > 0\}$. Then S_2 is unbounded. Suppose not, then there exists $M > 0$ such that $|z| \leq M$ for every $z \in S_2$. On taking $z = (M+1) \in S_2$ in particular, we see that $M+1 \leq M \implies 1 \leq 0$, a contradiction. Thus S_2 is unbounded.

Proposition 2.5. *Let S_1 and S_2 be two bounded sets. Then $S_1 \cup S_2$ and $S_1 \cap S_2$ are bounded sets.*

Proof. There exists $M_1 > 0, M_2 > 0$ such that $|s_1| \leq M_1$ and $|s_2| \leq M_2$ for all $s_1 \in M_1$ and $s_2 \in M_2$. If $s \in A \cup B$, then $|s| \leq \max(M_1, M_2)$, which proves the claim. The second part is easy and left as an exercise to the reader. □

2.4. Compact sets.

Definition 7. *Let $S \subset \mathbb{C}$. We say that S is compact if S is closed and bounded.*

Example 7. (1) Let $S_1 = \mathbb{C}$, then S_1 is not compact, as it is not bounded. It is however, closed.
(2) Let $S_2 = \{z \in \mathbb{C} : |z| < 1\}$, then S_2 is bounded and not closed. Hence, it is not compact.
(3) Let $S_3 = \{z \in \mathbb{C} : |z| \leq 1\}$. Then S_3 is compact.

Proposition 2.6. *Let S_1 and S_2 be two compact sets. Then $S_1 \cup S_2$ and $S_1 \cap S_2$ are compact sets.*

Proof. The proof follows from Proposition 2.3, Proposition 2.4 and Proposition 2.5. □

3. CONTINUITY

3.1. Sequences of complex numbers.

Definition 8. A sequence $(z_n)_{n \in \mathbb{N}}$ is a function $f : \mathbb{N} \rightarrow \mathbb{C}$ such that $f(n) = z_n$.

Definition 9. We say a sequence z_n converges to z as $n \rightarrow \infty$ if for every $\epsilon > 0$, there exists $n_0(\epsilon)$ such that $|z_n - z| < \epsilon$ for every $n \geq n_0(\epsilon)$. In notation, we write

$$\lim_{n \rightarrow \infty} z_n = z.$$

Example 8. (1) Let $z_n = \frac{1}{n}$. Let $\epsilon > 0$ be given. Let $n_0(\epsilon) = \lceil \frac{1}{\epsilon} \rceil + 1$, where $\lceil x \rceil$ denotes the greatest integer function at x . Let $n \geq n_0(\epsilon)$, Then

$$\left| \frac{1}{n} - 0 \right| = \left| \frac{1}{n} \right| = \frac{1}{n_0(\epsilon)} \leq \frac{1}{(1/\epsilon)} = \epsilon.$$

Hence

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0.$$

(2) Let $z_n = \frac{1}{n^2 + i}$. Let $\epsilon > 0$ be given. Let $n_0(\epsilon) = \lceil \frac{1}{\sqrt{\epsilon}} \rceil + 1$, where $\lceil x \rceil$ denotes the greatest integer function at x . Let $n \geq n_0(\epsilon)$, Then

$$\left| \frac{1}{n^2 + i} - 0 \right| = \left| \frac{1}{\sqrt{n^4 + 1}} \right| \leq \frac{1}{n^2} \leq \frac{1}{n_0^2(\epsilon)} \leq \frac{1}{(1/\epsilon)} = \epsilon.$$

Hence

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0.$$

Proposition 3.1. Let z_n be a sequence of complex numbers that is convergent. Then $\lim_{n \rightarrow \infty} z_n$ is unique.

Proof. Let $\lim_{n \rightarrow \infty} z_n = z$. Suppose $\lim_{n \rightarrow \infty} z_n = z'$. Let $\epsilon > 0$ be fixed. Then there exists $n_1(\epsilon) > 0$ and $n_2(\epsilon) > 0$ such that $|z_n - z| < \frac{\epsilon}{2}$ for $n \geq n_1(\epsilon)$ and $|z_n - z'| < \frac{\epsilon}{2}$ for $n \geq n_2(\epsilon)$. Using triangle inequality, for $n \geq n_1(\epsilon) + n_2(\epsilon)$,

$$|z - z'| \leq |z - z_n| + |z_n - z'| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

This shows that $z = z'$ as chosen ϵ is arbitrary. $\square$

Proposition 3.2. Let z_n be a sequence that is convergent. Then z_n is bounded, i.e., there exists $M > 0$ such that $|z_n| \leq M$ for all $n \in \mathbb{N}$.

The converse need not be true (see Example 9).

Proposition 3.3 (Algebra of limits of sequences). Let z_n and w_n be two sequences that converge to z and w , respectively. Then

- (1) $\lim_{n \rightarrow \infty} z_n + w_n = \lim_{n \rightarrow \infty} z_n + \lim_{n \rightarrow \infty} w_n = z + w$.
- (2) $\lim_{n \rightarrow \infty} z_n - w_n = \lim_{n \rightarrow \infty} z_n - \lim_{n \rightarrow \infty} w_n = z - w$.
- (3) $\lim_{n \rightarrow \infty} z_n w_n = (\lim_{n \rightarrow \infty} z_n)(\lim_{n \rightarrow \infty} w_n) = zw$.
- (4) $\lim_{n \rightarrow \infty} \frac{z_n}{w_n} = \frac{\lim_{n \rightarrow \infty} z_n}{\lim_{n \rightarrow \infty} w_n} = \frac{z}{w}$ provided $w_n \neq 0$ for all $n \in \mathbb{N}$ and $w \neq 0$.

Definition 10. Let z_n be a sequence of complex numbers. We say z_{n_k} is a subsequence of z_n if n_k is an increasing function when seen as a function of k defined on $\mathbb{N}$.

Example 9. (1) Let $z_n = \frac{1}{n+i}$ and $z_{2k} = \frac{1}{2k+i}$ and $z_{2k+1} = \frac{1}{2k+1+i}$. Then z_{2k} and z_{2k+1} are subsequences of z_n .
(2) Let $z_n = i^n$, then $z_{4k} = 1$ and $z_{4k+1} = i$. Note that $\lim_{k \rightarrow \infty} z_{4k} = 1$ and $\lim_{k \rightarrow \infty} z_{4k+1} = i$.

Proposition 3.4 (Sub-sequential criteria). *Let z_n be a sequence of complex numbers. Then, z_n converges to z if and only if for every subsequence z_{n_k} converges to z as $k \rightarrow \infty$.*

To demonstrate the above proposition, consider the sequence $z_n = i^n$. By Example 9(2) we see that $\lim_{k \rightarrow \infty} z_{4k} = 1$ and $\lim_{k \rightarrow \infty} z_{4k+1} = i$. Using Proposition 3.4, we conclude $1 = i$ a contradiction. Hence $\lim_{n \rightarrow \infty} i^n$ does not exist. Similarly, one can show that the sequence $\sin(\frac{n\pi}{2})$ does not exist.

3.2. Continuity.

Definition 11. *Let f be a complex valued function in a domain $D \subset \mathbb{C}$. let z_0 be a limit point of D . We say that $\lim_{z \rightarrow z_0} f(z) = L$ if for every $\epsilon > 0$ there exists $\delta(\epsilon) > 0$ such that $|f(z) - L| < \epsilon$ whenever $0 < |z - z_0| < \delta$.*

Proposition 3.5. *If $\lim_{z \rightarrow z_0} f(z) = L$ and $\lim_{z \rightarrow z_0} f(z) = L'$, then $L = L'$.*

Proof. The proof is similar to the proof of Proposition 3.1. □

Example 10. (1) $\lim_{z \rightarrow i} \bar{z} = -i$. To see this, let $\epsilon > 0$ be given. Let $\delta = \epsilon$ and $|z - i| < \delta$. Since $|\bar{z} + i| = |\bar{z} - \bar{i}| = |z - i| < \epsilon$, $\lim_{z \rightarrow i} \bar{z} = -i$.
(2) $\lim_{z \rightarrow 0} \frac{|z|}{z}$ does not exist. To see this let $z = x > 0$, then $\lim_{z \rightarrow 0} \frac{|z|}{z} = 1$. Now consider $z = ix$ for $x > 0$, then $\lim_{z \rightarrow 0} \frac{|z|}{z} = \frac{1}{i}$. Since the limit of a function at a point needs to be unique, the desired limit does not exist.

Definition 12. *Let f be a complex valued function in a domain $D \subset \mathbb{C}$. Let z_0 be a limit point of D . We say f is continuous at z_0 if $\lim_{z \rightarrow z_0} f(z) = f(z_0)$. We say f is continuous on D if f is continuous at every limit point of D .*

Theorem 2 (Sequential criteria of Limit of a function). *Let f be a complex valued function in a domain $D \subset \mathbb{C}$. Let z_0 be a limit point of D . Then $\lim_{z \rightarrow z_0} f(z) = L$ if and only if for every sequence z_n converging to z , the sequence $f(z_n)$ converges to L .*

The sequential criterion given by the theorem above is very useful. To see this, consider the following corollary.

Corollary 3.1. *Let f and g be two complex-valued functions defined on $D \subset \mathbb{C}$ having z_0 as a limit point for D . Let $\lim_{z \rightarrow z_0} f(z) = L$ and $\lim_{z \rightarrow z_0} g(z) = M$. Then*

- (1) $\lim_{z \rightarrow z_0} f(z) + g(z) = \lim_{z \rightarrow z_0} f(z) + \lim_{z \rightarrow z_0} g(z) = L + M$.
- (2) $\lim_{z \rightarrow z_0} f(z) - g(z) = \lim_{z \rightarrow z_0} f(z) - \lim_{z \rightarrow z_0} g(z) = L - M$.
- (3) $\lim_{z \rightarrow z_0} f(z)g(z) = (\lim_{z \rightarrow z_0} f(z))(\lim_{z \rightarrow z_0} g(z)) = LM$.
- (4) $\lim_{z \rightarrow z_0} \frac{f(z)}{g(z)} = \frac{\lim_{z \rightarrow z_0} f(z)}{\lim_{z \rightarrow z_0} g(z)} = \frac{L}{M}$, provided $g(z) \neq 0$ in a neighborhood of z_0 and $M \neq 0$.

Proof. We prove the first part only, rest is left as an exercise to the reader. Let z_n be a sequence converging to z_0 . By Theorem 2, we see that $f(z_n)$ and $g(z_n)$ converge to L and M respectively. Proposition 3.3 implies that $\lim_{n \rightarrow \infty} f(z_n) + g(z_n) = L + M$. Again using Theorem 2 we get

$$\lim_{z \rightarrow z_0} f(z) + g(z) = L + M.$$

□

Let $D \subset \mathbb{C}$ and $f : D \rightarrow \mathbb{C}$ be a function. The set $\text{range}(f)$ is defined to be

$$\text{range}(f) := \{f(z) : z \in D\}.$$

Proposition 3.6. *Let $D \subset \mathbb{C}$ and $f : D \rightarrow \mathbb{C}$ be a continuous function. If D is compact, then $\text{range}(D)$ is also compact.*

With the assumption of continuity, one may ask for a similar analogue of the above proposition for a different type of set. However, the answer is negative for open, closed, and bounded sets. To see consider $f : \mathbb{C} \rightarrow \mathbb{C}$ given by $f(z) = 1$ for all z . This gives a counter example to the expectation of $\text{range}(f)$ to be open under continuity. If we consider the function $g(z) = e^z$ for $z \in \mathbb{C}$, then $\text{range}(f) = \mathbb{C} \setminus \{0\}$ is not closed.

Theorem 3 (Topological definition of continuity I). *Let $D \subset \mathbb{C}$ and $f : D \rightarrow \mathbb{C}$ be a function. For a given subset S of $\mathbb{C}$, let $f^{-1}(S) = \{z \in D : f(z) \in S\}$ f is continuous on D if and only if for every open set $S \subset \mathbb{C}$, $f^{-1}(S) \cap D$ is an open set.*

An analog definition in terms of closed sets is given as follows.

Theorem 4 (Topological definition of continuity II). *Let $D \subset \mathbb{C}$ and $f : D \rightarrow \mathbb{C}$ be a function. For a given subset S of $\mathbb{C}$, let $f^{-1}(S) = \{z \in D : f(z) \in S\}$ f is continuous on D if and only if for every closed set $S \subset \mathbb{C}$, $f^{-1}(S) \cap D$ is an closed set.*

4. COMPLEX INTEGRATION

A curve $z(t) = x(t) + iy(t)$, $t \in [a, b]$, having a continuous derivative (i.e., $z(t)$ and $z'(t)$ are continuous on $[a, b]$) is said to be smooth or continuously differentiable on $[a, b]$. Although definition of smooth can more demanding like having derivative of any order, we will only need $z(t)$ and $z'(t)$ are continuous on $[a, b]$ for our purpose. By the derivatives at the end points a, b , we mean the appropriate one sided derivatives $z'(a+)$ and $z'(b-)$. For example,

$$z'(a+) = \lim_{t \rightarrow a^+} \frac{z(t) - z(a)}{t - a}.$$

A curve γ that is not smooth but consists of a finite sequence of smooth curves $\gamma_1, \gamma_2, \dots, \gamma_n$ joined together end-to-end is said to be a continuous piecewise smooth curve defined on a closed interval $[a, b]$. A curve having a sectionally continuous derivative is called a contour. In other words, a piecewise smooth curve is called a contour.

Definition 13. Let C be a piecewise smooth curve on $[a, b]$ and f a continuous function on the graph/trace of C .

(1) The contour integral of f along C is defined to be

$$\int_C f(z) dz = \int_a^b f(z(t)) z'(t) dt.$$

Sometimes the notation $\int_\gamma f(z) dz$ or $\int_\gamma f dz$ is used when γ is a piecewise smooth curve.

(2) We define

$$\int_C |f(z)| |dz| := \int_a^b |f(z(t))| |z'(t)| dt.$$

(3) The length of the curve C is defined to be

$$\int_C |dz| = \int_a^b |z'(t)| dt.$$

If the length of a curve is finite, then the curve is said to be rectifiable.

Theorem 5 (ML-Inequality). Suppose $f(z)$ is continuous on a contour C having length L , with $|f(z)| \leq M$ on C . Then

$$\left| \int_C f(z) dz \right| \leq \int_C |f(z)| |dz| \leq M \int_C |dz| = ML.$$

Proof. Since C is parametrized by $z(t)$ on the interval $[a, b]$, we have

$$\left| \int_C f(z) dz \right| = \left| \int_a^b f(z(t)) z'(t) dt \right| \leq \int_a^b |f(z(t))| |z'(t)| dt$$

since $|\int_a^b g(x) dx| \leq \int_a^b |g(x)| dx$ for a continuous function g on $[a, b]$. Similarly

$$\begin{aligned} & \int_a^b |f(z(t))| |z'(t)| dt \\ & \leq M \int_a^b |z'(t)| dt = ML \quad (\text{since } |f(z)| \leq M \text{ on } C) \end{aligned}$$

and the conclusion follows.

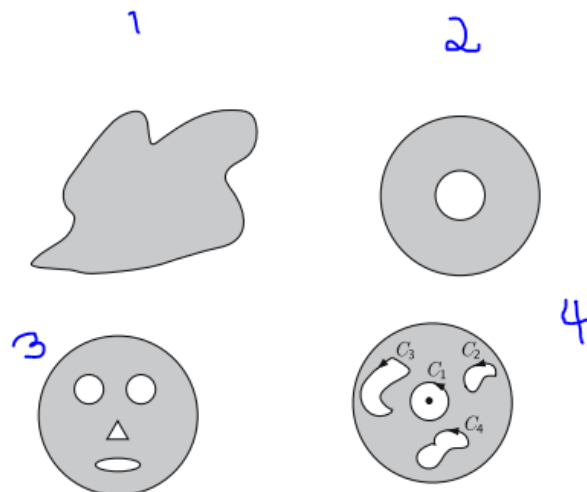


FIGURE 1. Illustration for simply connected and multiconnected domain

Theorem 6 (Cauchy’s theorem for a rectangle). *Let $f(z)$ be analytic in a domain containing a rectangle C and its interior. Then $\int_C f(z) dz = 0$.*

Theorem 7 (Cauchy’s theorem for a disk). *Let $f(z)$ be analytic in a domain containing the closed disk $|z - z_0| \leq r$. Then $\int_{|z-z_0|=r} f(z) dz = 0$.*

Definition 14. *If the initial and terminal points of a piecewise smooth curve C coincide, then C is said to be a closed curve. A domain D is simply connected if each simple closed curve contained in D contains only points of D inside.*

Geometrically, a “simply connected domain” has “no holes” inside, for if a simple closed curve should surround a hole, then the curve could not be shrunk beyond the hole. Here again, removal of a single point from a domain is akin to punching a hole in it. A domain that is not simply connected is said to be multiply connected. Open disks, open rectangles and star shaped domains are simple examples of simply connected domains. Punctured disks, punctured rectangles, and the punctured plane all have one “hole” and hence are not simply connected. For instance, in Figure 4, picture 1 is simply connected and picture 2,3,4 are multiconnected.

Theorem 8 (Cauchy’s Theorem). *If $f(z)$ is analytic in a simply connected domain D and C is a closed contour lying in D , then $\int_C f(z) dz = 0$.*

Theorem 9 (Cauchy’s Theorem for multiply connected domains). *Let D be a multiply connected domain bounded externally by a simple closed contour C and internally by n simple closed nonintersecting contours $C_1, C_2, \dots, C_n$. Let f be analytic on $D \cup C \cup C_1 \cup \dots \cup C_n$. Then*

$$\int_C f(z) dz = \sum_{k=1}^n \int_{C_k} f(z) dz$$

where C is taken counterclockwise around the external boundary C and clockwise around the internal boundaries $C_1, C_2, \dots, C_n$.

4.1. Cauchy integral formulae.

Theorem 10 (Cauchy's First Integral Formula). *Let f be analytic in a simply connected domain containing the simple closed contour C . If z_0 is inside C , then*

$$f(z_0) = \frac{1}{2\pi i} \int_C \frac{f(z)}{z - z_0} dz.$$

Proof. Since f is analytic in the domain, f is continuous on the domain. Hence, given $\epsilon > 0$, we can construct a circle $C_r : |z - z_0| = r$ inside C and small enough so that $|f(z) - f(z_0)| < \epsilon$ for all z on C_r . Using Theorem 9,

$$\int_C \frac{f(z)}{z - z_0} dz = \int_{C_r} \frac{f(z)}{z - z_0} dz.$$

On writing $f(z) = f(z_0) + (f(z) - f(z_0))$, one has

$$\begin{aligned} \int_C \frac{f(z)}{z - z_0} dz &= \int_{C_r} \frac{f(z_0)}{z - z_0} dz + \int_{C_r} \frac{f(z) - f(z_0)}{z - z_0} dz \\ &= 2\pi i f(z_0) + \int_{C_r} \frac{f(z) - f(z_0)}{z - z_0} dz, \quad \text{since } \int_{C_r} \frac{1}{z - z_0} dz = 2\pi i. \end{aligned}$$

Since the integral in the left side has a fixed value, as does $2\pi i f(z_0)$, it follows that for each C_r , the value of the contour integral

$$\int_{C_r} \frac{f(z) - f(z_0)}{z - z_0} dz$$

is constant. We now show that this value must be zero. Using Theorem 5,

$$\left| \int_{C_r} \frac{f(z) - f(z_0)}{z - z_0} dz \right| \leq \int_{C_r} \left| \frac{f(z) - f(z_0)}{z - z_0} \right| |dz| < \frac{\epsilon(2\pi r)}{r} = 2\pi\epsilon.$$

Since ϵ is arbitrary, the integral is zero. This concludes the proof.

Theorem 11 (Generalized Cauchy's Integral Formula). *Let f be analytic in a simply connected domain containing the simple closed contour C . Then f has derivatives of all orders at each point z_0 inside C , with*

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_C \frac{f(z)}{(z - z_0)^{n+1}} dz.$$

Theorem 12 (Taylor's Theorem). *Let $f(z)$ be analytic in a domain D whose boundary is C . If z_0 is a point in D , then $f(z)$ may be expressed as*

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n$$

and the series converges for $|z - z_0| < \delta$, where δ is the distance from z_0 to the nearest point on C .

Theorem 13. *Let $\{f_n\}$ be a sequence of functions continuous on a contour C , and suppose that $\{f_n\}$ converges uniformly to f on C . Then*

$$\lim_{n \rightarrow \infty} \int_C f_n(z) dz = \int_C \lim_{n \rightarrow \infty} f_n(z) dz = \int_C f(z) dz.$$

Corollary 4.0.1. Suppose $\{f_n\}$ be a sequence of continuous functions and that $\sum_{k=1}^n f_k$ converges uniformly on a contour C . Then

$$\sum_{n=0}^{\infty} \left(\int_C f_n(z) dz \right) = \int_C \left(\sum_{n=0}^{\infty} f_n(z) \right) dz.$$

Theorem 14 (Morera's Theorem). Let f be continuous in a domain D . If $\int_C f(z) dz = 0$ along every simple closed contour C contained in D , then $f(z)$ is analytic in D .

Corollary 4.0.2. Let f be analytic in a simply connected domain D . If z_0 is a point in D , then $F(z) := \int_{z_0}^z f(\zeta) d\zeta$ is analytic in D .

Theorem 15. Suppose $\{f_n\}$ converges uniformly to a function f on E . If each f_n is continuous at a point $z_0 \in E$, then the limit function f is also continuous at z_0 . That is,

$$(4.1) \quad \lim_{z \rightarrow z_0} \left(\lim_{n \rightarrow \infty} f_n(z) \right) = \lim_{n \rightarrow \infty} \left(\lim_{z \rightarrow z_0} f_n(z) \right).$$

Theorem 16. Let $\{f_n\}$ be a sequence of analytic functions converging uniformly to a function f on all compact subsets of a domain D . Then f is analytic in D .

Proof. It suffices to show that f is analytic at an arbitrary point z_0 in D . Consider a neighborhood D' of z_0 contained in D . By Theorem 15, f must be continuous at all points in D' . According to Theorem 13, along every simple closed contour C contained in D' ,

$$(4.2) \quad \lim_{n \rightarrow \infty} \int_C f_n(z) dz = \int_C f(z) dz.$$

Since f_n is analytic in D' , by Theorem 8,

$$\int_C f_n(z) dz = 0$$

for each n and each simple closed contour C . Using equation (4.2),

$$\int_C f(z) dz = 0.$$

By Morera's theorem, f must be analytic in D' . In particular, f is analytic at z_0 . This completes the proof.

Corollary 4.0.3. Suppose $\{f_n\}$ is a sequence of functions analytic in a domain D , and that $f = \sum_{n=0}^{\infty} f_n$, the series being uniformly convergent on all compact subsets of D . Then for all z in D ,

$$f'(z) = \sum_{n=0}^{\infty} f'_n(z).$$

Example 11. We wish to show $\sum_{n=1}^{\infty} 3^{-n} \sin(nz)$ represents an analytic function in the strip $|\operatorname{Im} z| < \ln 3$. Indeed,

$$\begin{aligned} |3^{-n} \sin(nz)| &= 3^{-n} \left| \frac{e^{inz} - e^{-inz}}{2i} \right| \\ &\leq \frac{3^{-n}}{2} (e^{n|\operatorname{Im} z|} + e^{-n|\operatorname{Im} z|}) = e^{-n(\ln 3 - |\operatorname{Im} z|)}. \end{aligned}$$

The Weierstrass M -test shows that $\sum_{n=1}^{\infty} 3^{-n} \sin(nz)$ converges uniformly on each compact subset of $D = \{z : |\operatorname{Im} z| < \ln 3\}$. By Corollary 4.0.3, the given series of functions represents an analytic function for $|\operatorname{Im} z| < \ln 3$.

4.2. Applications of Cauchy theorem.

Theorem 17 (Cauchy's Inequality). *Suppose f is analytic inside and on the circle C having centre at z_0 and radius r . If $|f(z)| \leq M$ on C , then*

$$|f^{(n)}(z_0)| \leq \frac{n!M}{r^n}, \quad n = 1, 2, \dots$$

Proof. By the generalised Cauchy integral formula (Theorem 11),

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_C \frac{f(z)}{(z - z_0)^{n+1}} dz.$$

Hence, by the hypothesis and ML inequality (Theorem 5),

$$|f^{(n)}(z_0)| \leq \frac{n!}{2\pi} \int_C \frac{|f(z)|}{|z - z_0|^{n+1}} |dz| \leq \frac{n!M}{2\pi} \int_C \frac{|dz|}{|z - z_0|^{n+1}}.$$

Since $|z - z_0| = r$,

$$\frac{n!M}{2\pi} \int_C \frac{|dz|}{|z - z_0|^{n+1}} = \frac{n!M}{2\pi} \int_C \frac{|dz|}{r^{n+1}} = \frac{n!M}{r^n}$$

and the proof is complete. $\square$

An easy corollary to the above theorem is as follows.

Corollary 4.0.4. *Suppose that f is analytic for $|z - z_0| < r$ for some $r > 0$. If $|f(z) - b| \leq M$ for $|z - z_0| < r$ and $b \in \mathbb{C}$, then $|f'(z_0)| \leq \frac{M}{r}$.*

Proof. Take $0 < \delta < r$. Then $g(z) = f(z) - b$ is analytic and maps the disk $|z - z_0| \leq \delta$ into the set $\{w \in \mathbb{C} : |w| \leq M\}$. Hence by Cauchy's inequality, on taking z_0 and radius equal to δ ,

$$|f'(z_0)| = |g'(z_0)| \leq \frac{M}{\delta}.$$

Letting $\delta \rightarrow r$ proves the result. $\square$

As a comparison with real analysis, we consider $f(x) = \cos(1/x)$ for $x > 0$. Then f is differentiable for $x > 0$ and $|f(x)| \leq 1$ for all $x > 0$. However,

$$f'(x) = \frac{1}{x^2} \sin(1/x)$$

which is clearly not bounded. To see this for each $n \in \mathbb{N}$, consider

$$\left| f' \left(\frac{1}{n\pi + \frac{\pi}{2}} \right) \right| = \left(n\pi + \frac{\pi}{2} \right)^2.$$

As $n \rightarrow \infty$, $\left| f' \left(\frac{1}{n\pi + \frac{\pi}{2}} \right) \right| \rightarrow \infty$ showing $\{|f'(x)| : x > 0\}$ being unbounded. However, Corollary 4.0.4 tells us that the derivative should be bounded.

Definition 15. A function $f : \mathbb{C} \rightarrow \mathbb{C}$ is said to be entire if f is analytic on $\mathbb{C}$.

Theorem 18 (Liouville's Theorem). *A bounded entire function must be a constant function.*

Proof. Suppose f is entire with $|f(z)| \leq M$ for all z . Given any complex number z_0 , we have by Cauchy's inequality, $|f'(z_0)| \leq M/r$ for every positive number r . Letting $r \rightarrow \infty$, we deduce that $f'(z_0) = 0$. Since z_0 was arbitrary, $f'(z) = 0$ for all z , and hence f is constant. $\square$

Proof. Suppose f is entire, then $|f(z)| \leq M$ for all z . Given any complex number z_0 , we have by Cauchy's inequality, $|f'(z_0)| \leq M/r$ for every positive number r . Letting $r \rightarrow \infty$, we deduce that $f'(z_0) = 0$. Since z_0 was arbitrary, $f'(z) = 0$ for all z , and hence f is constant. $\square$

As an application of Liouville's theorem, we prove the fundamental theorem of algebra.

Theorem 19 (The fundamental theorem of algebra). *Every nonconstant polynomial has at least one zero.*

Proof. Consider $P(z) = a_0 + a_1z + \cdots + a_nz^n$, $a_n \neq 0$. If $P(z)$ never vanishes, then $1/P(z)$ is entire. Furthermore $1/P(z) \rightarrow 0$ as $z \rightarrow \infty$. Thus there exists $R > 0$ such that $|1/P(z)| \leq 1$ for $|z| > R$. But $1/P(z)$ is continuous on the compact set $|z| \leq R$. Using Proposition 3.6, the set $\{f(z) : |z| \leq R\}$ is compact and hence bounded. Therefore, $1/P(z)$ is bounded in $\mathbb{C}$ and, by Liouville's theorem, must be a constant. This implies that $P(z)$ is also a constant, contradicting our assumption. $\square$

Corollary 4.0.5. *Every polynomial of degree n has exactly n (not necessarily distinct) zeros.*

Proof. The fundamental theorem shows the existence of at least one zero, say r_1 for a degree n polynomial $P(z)$. The expression $z - r_1$ can be factored out, leaving a polynomial of degree $n - 1$, namely $\frac{P(z)}{(z - r_1)}$. Reapplying the theorem to this new polynomial, we obtain another zero. This process can be repeated n times. $\square$

Theorem 20 (Uniqueness Theorem). *Let f be analytic in a domain D , and let $\{z_n\}$ is a sequence of distinct points converging to a point z_0 in D . If $f(z_n) = 0$ for each n , then $f(z) \equiv 0$ throughout D .*

Theorem 21 (Identity Theorem). *Suppose $\{z_n\}$ is a sequence of points having a limit point in a domain D . If $f(z)$ and $g(z)$ are analytic in D , with $f(z_n) = g(z_n)$ for each n , then $f = g$ throughout D .*

Proof. Let z_0 denote the limit point of $\{z_n\}$. By Theorem 3.4, there exists a subsequence $\{z_{n_k}\}$ converging to z_0 . Setting $h(z) = f(z) - g(z)$, we see that $h(z_{n_k}) = 0$ for all points of the sequence $\{z_{n_k}\}$. Applying Theorem 21 for h , we have $h(z) = 0$ for all $z \in D$ and the result follows. $\square$

Remark 1. *Cauchy's integral formula tells that the behaviour of an analytic function on a simple closed contour determines its behaviour inside. The identity theorem tells us even more. It says that the behaviour at any sequence of points, inside or on the simple closed contour, determines the behaviour of the analytic function at all points of the domain.*

Remark 2. *We now have a simple way to prove the standard trigonometric identities using Theorem 21. For example, the function*

$$f(z) = \sin 3z - 3 \sin z + 4 \sin^3 z$$

is an entire function that is equal to 0 on the real axis. Hence $f(z) = 0$ in the complex plane, that is, $\sin 3z = 3 \sin z - 4 \sin^3 z$ for all z .

5. LAURENT SERIES AND RESIDUE THEOREM

Theorem 22 (Laurent's theorem). *Let f be analytic in the annulus $R_1 < |z - z_0| < R_2$. Then the representation*

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$$

is valid throughout the annulus. Furthermore, the coefficients a_n are given by

$$a_n = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta \quad (n = 0, \pm 1, \pm 2, \dots),$$

where C is any simple closed contour contained in the annulus that makes a complete counterclockwise revolution about the point z_0 .

The series of negative powers of $(z - z_0)$ is called the **principal part** of the Laurent expansion, while the series of positive powers is called the **analytic part**.

Example 12. *To find the Laurent expansion for*

$$f(z) = \frac{\sin z}{z^2} \quad (|z| > 0),$$

we expand $\sin z$ in a Maclaurin series. This observation leads to

$$f(z) = \frac{\sin z}{z^2} = \frac{1}{z^2} \left(z - \frac{z^3}{3!} + \frac{z^5}{5!} - \dots \right) = \frac{1}{z} - \frac{z}{3!} + \frac{z^3}{5!} - \dots \quad (|z| > 0).$$

Similarly, from the identity $e^w = 1 + w + w^2/2! + w^3/3! + \dots$, we get

$$f(z) = e^{1/z} = 1 + \frac{1}{z} + \frac{1}{2!} \frac{1}{z^2} + \frac{1}{3!} \frac{1}{z^3} + \dots \quad (|z| > 0).$$

Example 13. *Consider the function*

$$f(z) = \frac{1}{z^2 + 1} = \frac{1}{(z + i)(z - i)}$$

which is analytic in $\mathbb{C} \setminus \{\pm i\}$. We expand this function in a Laurent series valid in a deleted neighborhood of $z = i$. To do this, we consider

$$\frac{1}{z + i} = \frac{1}{2i} + \frac{z - i}{2i} = \frac{1}{2i} \left[1 + \frac{z - i}{2i} \right] = \frac{1}{2i} \sum_{n=0}^{\infty} (-1)^n \left(\frac{z - i}{2i} \right)^n$$

which is valid for $|z - i| < 2$. Hence, for $0 < |z - i| < 2$, we have

$$\begin{aligned} f(z) &= \frac{1}{(z + i)(z - i)} = \frac{1}{2i(z - i)} \sum_{n=0}^{\infty} (-1)^n \left(\frac{z - i}{2i} \right)^n = - \sum_{n=1}^{\infty} \left(-\frac{1}{2i} \right)^n (z - i)^{n-2} \\ &= - \sum_{n=-1}^{\infty} \left(\frac{i}{2} \right)^n (z - i)^n. \end{aligned}$$

In the Laurent series expansion (see Theorem 22), if $R_1 = 0$, then the point z_0 becomes an isolated singularity of $f(z)$. In view of this, the Laurent series has an isolated singularity at z_0 . Suppose f possesses a Laurent series expansion

$$f(z) = \sum_{n=-\infty}^{-1} a_n(z - z_0)^n + \sum_{n=0}^{\infty} a_n(z - z_0)^n$$

valid for $0 < |z - z_0| < \delta$ for some $\delta > 0$. Then the following situations arise.

- (1) No principal part: In this case $a_n = 0$ for all $n < 0$. So the above Laurent series simply reduces to

$$f(z) = a_0 + a_1(z - z_0) + a_2(z - z_0)^2 + \cdots, \quad 0 < |z - z_0| < \delta.$$

It follows that $\lim_{z \rightarrow z_0} f(z) = a_0$, i.e., f has a removable singularity at z_0 if and only if $a_n = 0$ for all $n < 0$. For instance, take $f(z) = \frac{\sin z}{z}$ for $z \neq 0$ and $z_0 = 0$.

- (2) The principal part consists of a finite number of terms: In this case, $a_n = 0$ for all $n < -m$ and $a_{-m} \neq 0$ for some $m \geq 1$. So the Laurent series about z_0 reduces to

$$f(z) = \frac{a_{-m}}{(z - z_0)^m} + \cdots + \frac{a_{-1}}{z - z_0} + \sum_{n=0}^{\infty} a_n(z - z_0)^n.$$

which is valid for $0 < |z - z_0| < \delta$ so that

$$\lim_{z \rightarrow z_0} (z - z_0)^m f(z) = a_{-m} \neq 0.$$

In other words, f has a pole of order m at z_0 if and only if there exists an $m \in \mathbb{N}$ such that $a_{-m} \neq 0$ and $a_n = 0$ for all $n < -m$. For instance, take $f(z) = \frac{\cos z^2}{z^8}$ for $z \neq 0$ and $z_0 = 0$.

- (3) The principal part consists of infinitely many terms: In this case, $a_n \neq 0$ for infinitely many negative integer values of n . Thus the principal part has infinitely many terms if and only if the singularity at z_0 is an essential singularity. For instance, take $f(z) = e^{\frac{1}{z}}$ for $z \neq 0$ and $z_0 = 0$.

If f is analytic in a deleted neighborhood of z_0 , then by Laurent's theorem we may write

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n \quad (0 < |z - z_0| < \delta),$$

where

$$a_n = \frac{1}{2\pi i} \int_C \frac{f(z)}{(z - z_0)^{n+1}} dz \quad (n \in \mathbb{Z}).$$

Here C is any simple closed contour enclosing z_0 and contained in the neighborhood. In particular,

$$(5.1) \quad a_{-1} = \frac{1}{2\pi i} \int_C f(z) dz, \quad \text{i.e.,} \quad \int_C f(z) dz = 2\pi i a_{-1}.$$

Equation (5.1) says that evaluating a certain integral of $f(z)$ around C that encloses no other singularity other than z_0 is akin to determining a certain coefficient in Laurent series. Therefore, if we can compute a_{-1} , then we can find $\int_C f(z) dz$. The coefficient a_{-1} is called the residue of $f(z)$ at z_0 and is denoted by

$$\text{Res}[f(z); z_0].$$

Example 14. (1) As $e^{1/z} = 1 + 1/z + 1/(2!z^2) + \cdots$ for $|z| > 0$, $\text{Res}[e^{1/z}; 0] = 1$ and so

$$\int_{|z|=1} e^{1/z} dz = 2\pi i.$$

More generally,

$$\int_C \exp\left(\frac{1}{z^k}\right) dz = \begin{cases} 0 & \text{if } k \neq 1, k \in \mathbb{Z}, \\ 2\pi i & \text{if } k = 1, k \in \mathbb{Z}, \end{cases}$$

where C is a simple closed contour enclosing the origin.

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